STA2002 Tutorial 4

1 Review

1.1 The Neyman-Pearson paradigm

Let $X_1, \ldots, X_n \overset{i.i.d.}{\sim}$ from a common distribution with unknown parameter θ. We have an initial belief on the values that θ can take, and we call this **null hypothesis** H_0. We would like to test this belief against a competing belief, and we call this **alternative hypothesis** H_1.
Once we formulate H_0 and H_1, we collect sample data $x_1, \ldots, x_n$ to see whether H_0 or H_1 is favored.

Test Statistics (T): the statistics computed from a sample, based on which the decision is made.
Critical region (C): the set of values of the test statistics that leads to rejection of H_0.

	H_0 true	H_1 true
Reject H_0	**Type I error**	Correct
Fail to reject H_0	Correct	**Type II error**

Significance level: $\alpha = P(\text{Test makes Type I Error}) = P(T \in C; H_0)$

Power: $1 - \beta = 1 - P(\text{Test makes Type II Error}) = 1 - P(T \notin C; H_1)$

By changing C, α increases (decreases) while β decreases (increases). Both errors cannot be reduced at the same time.

Uniformly most powerful test: The test with minimal type II error β for every θ under H_1.
The Neyman-Pearson Lemma: Consider the simple hypothesis test H_0: $\theta = \theta_0$ versus $H_1 : \theta = \theta_1$.
a). Fix type I error α.
b). The test (with a cut-off for the likelihood ratio):

$$\text{Rejecting } H_0 \text{ if } \frac{L(\theta_0|x)}{L(\theta_1|x)} \leq k$$

and

$$\text{Fail to reject } H_0 \text{ if } \frac{L(\theta_0|x)}{L(\theta_1|x)} > k$$

is the uniformly most powerful test (with minimized β).

p-value: the probability of observing a more extreme test statistic T (in the direction that favors H_1), given that H_0 is true. **Reject H_0 if p-value $\leq \alpha$.**

1.2 Tests for mean and variance

One sample test of mean Suppose $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$. σ^2 is unknown. $H_0 : \mu = \mu_0$
Test statistic

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t(n-1) \text{ under } H_0$$

Compute $t = \frac{\bar{x} - \mu_0}{S/\sqrt{n}}$.
a). $H_0 : \mu = \mu_0$, $H_1 : \mu > \mu_0$. Reject H_0 when:

$$\boxed{\text{p-value} = P(t(n-1) > t) \leq \alpha} \Leftrightarrow \boxed{t \geq t_\alpha(n-1)}$$

b). $H_0 : \mu = \mu_0$, $H_1 : \mu < \mu_0$. Reject H_0 when:

$$\boxed{\text{p-value} = P(t(n-1) < t) \leq \alpha} \Leftrightarrow \boxed{t \leq -t_\alpha(n-1)}$$

c). $H_0 : \mu = \mu_0$, $H_1 : \mu \neq \mu_0$. Reject H_0 when:

$$\boxed{\text{p-value} = 2P(t(n-1) > |t|) \leq \alpha} \Leftrightarrow \boxed{|t| \geq t_{\alpha/2}(n-1)}$$

One sample test of variance Suppose $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, test $H_0 : \sigma^2 = \sigma_0^2$ versus $H_1 : \sigma^2 \neq \sigma_0^2$
Test statistic

$$T = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2_{n-1} \text{ under } H_0$$

Reject H_0 when:

$$\boxed{T \leq \chi^2_{n-1,1-\alpha/2} \text{ or } T \geq \chi^2_{n-1,\alpha/2}}$$

Two sample test of means Suppose $X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$, $Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$. $H_0 : \mu_X = \mu_Y$ versus $H_1 : \mu_X \neq \mu_Y$
1). $\sigma_X^2 = \sigma_Y^2 = \sigma^2$, σ^2 is unknown. Test statistics

$$T = \frac{\bar{X} - \bar{Y}}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t(n+m-2) \text{ under } H_0$$

where $S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}$. Reject H_0 when:

$$\boxed{\text{p-value} = 2P(t(n+m-2) > |t|) \leq \alpha} \Leftrightarrow \boxed{|t| \geq t_{\alpha/2}(n+m-2)}$$

2). $\sigma_X^2 \neq \sigma_Y^2$, both are unknown. Test statistics

$$T = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}} \sim t(r) \text{ approximately, under } H_0$$

where $r = \left\lfloor \frac{(\frac{S_X^2}{n} + \frac{S_Y^2}{m})^2}{\frac{1}{n-1}(\frac{S_X^2}{n})^2 + \frac{1}{m-1}(\frac{S_Y^2}{m})^2} \right\rfloor$. Reject H_0 when:

$$\boxed{\text{p-value} = 2P(t(r) > |t|) \leq \alpha} \Leftrightarrow \boxed{|t| \geq t_{\alpha/2}(r)}$$

Two sample test of variances Suppose $X_1, \ldots, X_n \sim N(\mu_X, \sigma_X^2)$, $Y_1, \ldots, Y_m \sim N(\mu_Y, \sigma_Y^2)$. $H_0 : \sigma_X^2 = \sigma_Y^2$ versus $H_1 : \sigma_X^2 \neq \sigma_Y^2$
Test statistic

$$T = \frac{S_X^2}{S_Y^2} \sim F(n-1, m-1) \text{ under } H_0$$

Reject H_0 at significant level α when

$$\boxed{T \leq F_{1-\alpha/2}(n-1, m-1) \text{ or } T \geq F_{\alpha/2}(n-1, m-1)}$$

Note that

$$F_{1-\alpha/2}(n-1, m-1) = \frac{1}{F_{\alpha/2}(m-1, n-1)}$$

1. We assume that $X_1, X_2, \cdots, X_n$ are samples from $N(\mu, 1)$, and consider the following hypothesis test problems:

$$H_0 : \mu = 2 \quad \text{vs} \quad H_1 : \mu = 3$$

If the test is determined by the rejection region $W = \{\bar{x} \geqslant 2.6\}$

(a) Please find the probabilities of Type I error and Type II error when $n = 20$.

(b) If we require the probability of Type II error $\beta \leqslant 0.01$, please find the least value n should be.

(c) Please prove that when $n \to \infty, \alpha \to 0$ and $\beta \to 0$.

(a) From the definition, we know the probability of Type I error is:

$$\alpha = P\left(\overline{\boldsymbol{x}} \geqslant 2.6 \mid H_0\right) = P\left(\frac{\bar{x} - 2}{\sqrt{1/20}} \geqslant \frac{2.6 - 2}{\sqrt{1/20}}\right) = 1 - \Phi(2.68) = 0.0037$$

This is because when H_0 holds, $\overline{\boldsymbol{x}} \sim \mathcal{N}(2, 1/20)$. And the probability of Type II error is

$$\begin{aligned}\beta &= P\left(\bar{x} < 2.6 \mid H_1\right) = P\left(\frac{\bar{x} - 3}{\sqrt{1/20}} < \frac{2.6 - 3}{\sqrt{1/20}}\right) = \Phi(-1.79) \\ &= 1 - \Phi(1.79) = 0.0367\end{aligned}$$

This is because when H_1 holds, $\bar{x} \sim \mathcal{N}(3, 1/20)$.

(b) If we need to require that the probability of Type II error satisfies

$$\beta = P\left(\bar{x} < 2.6 \mid H_1\right) = P\left(\frac{\bar{x} - 3}{\sqrt{1/n}} < \frac{2.6 - 3}{\sqrt{1/n}}\right) \leqslant 0.01$$

which means $1 - \Phi\left(\frac{0.4}{\sqrt{1/n}}\right) \leqslant 0.01$. By referring to the quantile table, we have $0.4\sqrt{n} \geqslant 2.33$, and thus we get $n \geqslant 33.93$. In other word, the least value of n should be 34 to promise the probability of Type II error $\beta \leqslant 0.01$.

(c) when $n \to \infty, \Phi(0.6\sqrt{n}) \to 1$, which means $\alpha \to 0$. The probability of Type II error is:

$$\beta = P\left(\bar{x} < 2.6 \mid H_1\right) = P\left(\frac{\bar{x} - 2}{\sqrt{1/n}} < \frac{2.6 - 3}{\sqrt{1/n}}\right) = \Phi(-0.4\sqrt{n}) = 1 - \Phi(0.4\sqrt{n})$$

when $n \to \infty, \Phi(0.4\sqrt{n}) \to 1$, which means $\beta \to 0$.

2. Assume that the weight X in ounces of a "10-ounce" box of cornakes is $N(\mu, 0.03)$. Let $X_1, X_2, \cdots, X_n$ be a random sample from this distribution.
(You can skip (a) since the lecture has not covered LRT.)

   (a) To test the hypothesis $H_0 : \mu \geq 10.35$ against the alternative hypothesis $\mathrm{H}_1 : \mu < 10.35$, what is the critical region of size $\alpha = 0.05$ specied by the likelihood ratio test criterion?
   Hint: The LRT
   $$\lambda = \frac{\sup_{\theta\in\Theta_0}\ L(X_1,\ldots,X_n;\theta)}{\sup_{\theta\in\Theta}\ L(X_1,\ldots,X_n;\theta)} \leq k$$

   (b) If a random sample of $n = 50$ boxes yielded a sample mean of $\bar{x} = 10.31$, is H_0 rejected?

   (c) What is the p-value of this test?

   (a) Recall the generalized LRT:
   $$\lambda = \frac{\sup_{\theta\in\Theta_0}\ L(X_1,\ldots,X_n;\theta)}{\sup_{\theta\in\Theta}\ L(X_1,\ldots,X_n;\theta)} \leq k$$
   for this problem, $\Theta_0 = \{\mu|\mu \geq 10.35\}$, $\Theta = \{\mu|\mu \in \mathbb{R}\}$, and we have
   $$\lambda = \frac{[1/(\sqrt{0.06\pi})]^{n/2}\exp\left[-\sum_{i=1}^n (x_i - \hat{\mu}_0)^2/(0.06)\right]}{[1/(\sqrt{0.06\pi})]^{n/2}\exp\left[-\sum_{i=1}^n (x_i - \bar{x})^2/(0.06)\right]} \leq k \quad \hat{\mu}_0 \triangleq \underset{\mu\geq 10.35}{\operatorname{argmax}} L(X_1,\ldots,X_n,\mu)$$
   $$\Leftrightarrow \exp\left[-\frac{n}{0.06}(\bar{x} - \hat{\mu}_0)^2\right] \leq k$$
   $$\Leftrightarrow -\frac{n}{0.06}(\bar{x} - \hat{\mu}_0)^2 \leq \ln k$$

   - If $\bar{x} \geq 10.35, \hat{\mu}_0 = \bar{x}$.
     $\lambda = 1$, we can not reject H_0.
   - If $\bar{x} < 10.35$, then $\hat{\mu}_0 = 10.35$.
     $P\left(\frac{\bar{x}-10.35}{\sqrt{0.03/n}} \leq -\sqrt{-2\ln k}\right) = 0.05 \Rightarrow \leq -z_{0.05} = -1.645$

   (b)
   $$\frac{10.31 - 10.35}{\sqrt{0.03/50}} = -1.633 > -1.645$$
   do not reject H_0

   (c)
   $$P(Z \leq -1.633) = 0.0516$$

3. We now have measured the resistance (measuring unit: Ω) of two batches of electronic components samples are as follows:

Batch A (x)	0.140	0.138	0.143	0.142	0.144	0.137
Batch B (y)	0.135	0.140	0.142	0.136	0.138	0.140

Let us assume that the resistance values of these two samples obey $N(\mu_1, \sigma_1^2)$, $N(\mu_2, \sigma_2^2)$ respectively, and the two samples are independent.

(a) Please try to test whether the variances of these two populations are equivalent. (Take $\alpha = 0.05$)

(b) Based on the result in (a), please try to test whether the mean value of these two populations are equivalent. (Take $\alpha = 0.05$)

(a) We should do F test to verify whether the variances of two population are equivalent. Here we have:

$$\bar{x} = 0.1407,\ \bar{y} = 0.1385,\ s_x = 0.0028,\ s_y = 0.0027$$

If we take $\alpha = 0.05$, then we have $F_{0.975}(5,5) = 7.15$, $F_{0.025}(5,5) = \frac{1}{F_{0.975}(5,5)} = 0.1399$. Here we have the rejection region $W = \{F \leq 0.1399 \text{ or } F \geq 7.15\}$, while

$$F = \frac{s_x^2}{s_y^2} = \frac{0.0028^2}{0.0027^2} = 1.0754$$

As F-value doesn't belong to the rejection region, we accept the hypothesis that the variances of these two population are equivalent.

(b) Because we have accept the hypothesis in (a). that the variances of two populations are equivalent, we can do t test on the mean value test. When $\alpha = 0.05$, $t_{0.975}(10) = 2.2281$, so we have the rejection region $W = |t| \geq 2.2281$. Here we have:

$$s_p = \left(\frac{5 \times 0.0028^2 + 5 \times 0.0027^2}{6 + 6 - 2}\right)^{1/2} = 0.00275$$

$$t = \frac{0.1407 - 0.1385}{0.00275\sqrt{\frac{1}{6} + \frac{1}{6}}} = 1.3856 < 2.2281$$

So we accept H_0, which means the mean value of these two populations are equivalent.